# What a Study Can Prove KEY

Name _____

Two questions decide what a study can tell us.

1. Was the sample chosen at **random** from the population? If yes, we can **generalize** to the population.
2. Were the treatments given by **random assignment**? If yes, we can say the treatment **caused** the result.

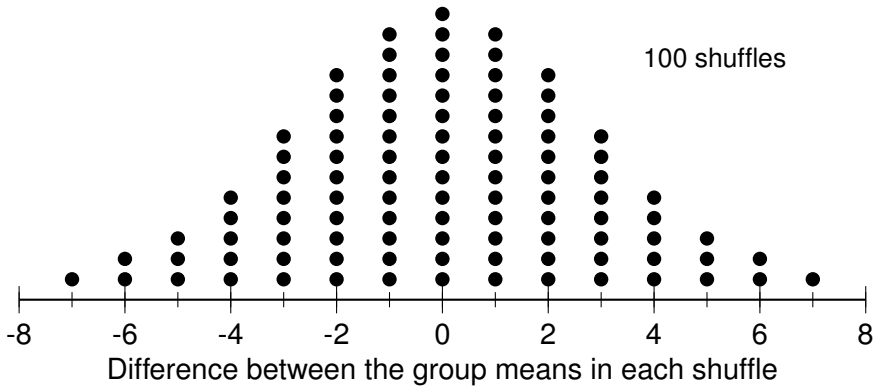
| | **Random assignment** | **No random assignment** |
|---|---|---|
| **Random sample** | Cause: yes<br>Generalize: yes | Cause: no<br>Generalize: yes |
| **Not a random sample** | Cause: yes, for these subjects<br>Generalize: no | Cause: no<br>Generalize: no |

**Observational studies** have no random assignment. They can show an **association**, but not a cause.

**Is the difference real, or just chance?**

20 students were randomly split into two groups of 10. One group took a 5-minute stretch break before a quiz. The break group scored **6 points higher** on average.

Even if the break does nothing, two random groups will score a little differently. To see how big a difference chance can make, we **simulate**: shuffle the 20 scores into two random groups of 10, find the difference, and repeat 100 times.




Only **3** of the 100 shuffles gave a difference of 6 or more. A 6-point difference is very unlikely to happen by chance alone. We say the result is **statistically significant**. The break probably helped.

**Rule of thumb:** if chance gives a result this big less than about 5% of the time, call it statistically significant.

Work with a partner. Compare answers when you finish each problem.

1. For each study, can we conclude cause? Can we generalize to the population? Explain.

   (a) Researchers randomly choose 1,000 adults in Missouri and ask how much they exercise and how often they get sick.

   Generalize to Missouri adults: yes (random sample). Cause: no (observational, no random assignment).

   (b) 40 volunteers are randomly assigned to drink green tea or water every day for a month. The tea group has lower blood pressure.

   Cause: yes, for these volunteers (random assignment). Generalize: no (volunteers, not a random sample).

   (c) A teacher compares the grades of students who chose to join a study club with students who did not.

   Cause: no (students chose). Generalize: no (not a random sample).

2. Use the simulation dot plot on page 1. Each dot is the difference between two random groups when the treatment does nothing.

   (a) In a plant experiment, the fertilizer group grew 2 cm more than the water group. How many of the 100 shuffles gave a difference of 2 or more?

   $11 + 8 + 5 + 3 + 2 + 1 = 30$

   (b) Is a 2 cm difference statistically significant? Explain.

   No. Chance alone gave a difference this big 30% of the time. That is common.

3. Find the mistake in each conclusion.

   (a) A survey of 500 randomly chosen adults found that people who drink coffee live longer. Conclusion: “Coffee makes you live longer.”

   This is an observational study. It shows an association, not a cause. Coffee drinkers may differ in other ways (lurking variables).

   (b) 30 runners from one gym were randomly assigned to two training plans. Plan A runners got faster. Conclusion: “Plan A will make every runner in the country faster.”

   The runners were not a random sample of all runners. We can only say Plan A caused the change for these runners.

   (c) In a randomized experiment, Group A scored 1 point higher than Group B. A simulation shows chance gives a difference of 1 or more about 40% of the time. Conclusion: “Treatment A works better.”

   The difference is not statistically significant. A 1-point difference happens by chance 40% of the time.

4. A headline says: “Students who eat dinner with their family get better grades.” The data came from a survey. Rewrite the headline so it does not claim a cause.

   “Students who eat dinner with their family tend to have better grades.” (Or: “Family dinners linked to better grades.”)